The matte sea

What `water_model: matte` does, why it exists, how well it works and how to set it. A reference for people running the playground; November 2026.

In short

- The matte sea paints the sea with its measured colour: a **flat, diffuse surface** whose albedo is π × Rrs(λ) + an offset.
- It exists because SMART-G's 3D mode (needed for clouds) cannot hold any water body, so the full sea model cannot be used with a cloud. A flat diffuse surface is accepted by the 3D mode.
- Cloud-free, away from the sun glint, it matches the full model to about ±2.5% in the three MODIS bands. In the glint it fails.
- Under a cloud the right **offset** depends on how bright the sky is; one number cannot suit a whole cumulus scene. Treat cloud-scene results as indicative.
- It costs about the same as a plain reflecting sea.

1 Why a matte sea

The sea has two kinds of light. **Reflected** light bounces off the wavy surface (the sun glint and the mirror-like sky reflection). **Water-leaving** light enters the water, is scattered by it and comes back up; it carries the sea's blue or green colour. The playground's full models simulate the second kind with a water object under a rough surface. SMART-G's 3D mode, the one that handles clouds, fails with any water object (`KeyError: 'iabs_oc'`; tested with the installed 2.0.0b1 and with 2.0.6, for the chlorophyll column and for the measured-spectrum reflector). With a cloud, the sea can only reflect, and looks too dark and colourless.

The matte sea keeps what the 3D mode does accept, a Lambertian surface (`LambSurface`), and gives it the sea's colour.

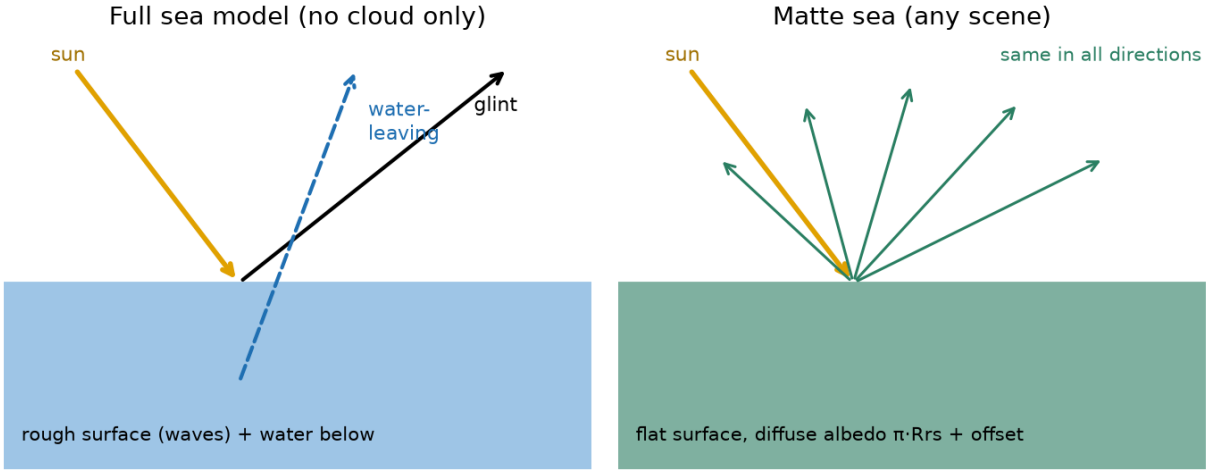


Figure 1. Left: the full model, a rough surface that reflects (glint) plus water that sends colour back up. Right: the matte sea, one flat surface that sends light back equally in every direction.

2 How it is built

The albedo. The input is the measured remote-sensing reflectance Rrs(λ), the water-leaving radiance divided by the irradiance on the sea (default: very clear water measured by the MOBY buoy; the greener NOMAD spectra can be chosen with `sea_spectrum_file`). A Lambertian surface of albedo A under irradiance E sends the radiance A E / π in every direction, so choosing **A = π × Rrs** reproduces the measured water-leaving radiance.

The offset. A real sea also reflects some sky light, which π × Rrs does not contain. The offset, added to the albedo, stands for it: A(λ) = π × Rrs(λ) + offset. Left empty (`matte_offset: null`), the playground uses values fitted for a clear sky: 0.0104, 0.0047 and 0.0029 at 469, 555 and 645 nm (interpolated between, held constant outside). A number you set is used at every wavelength.

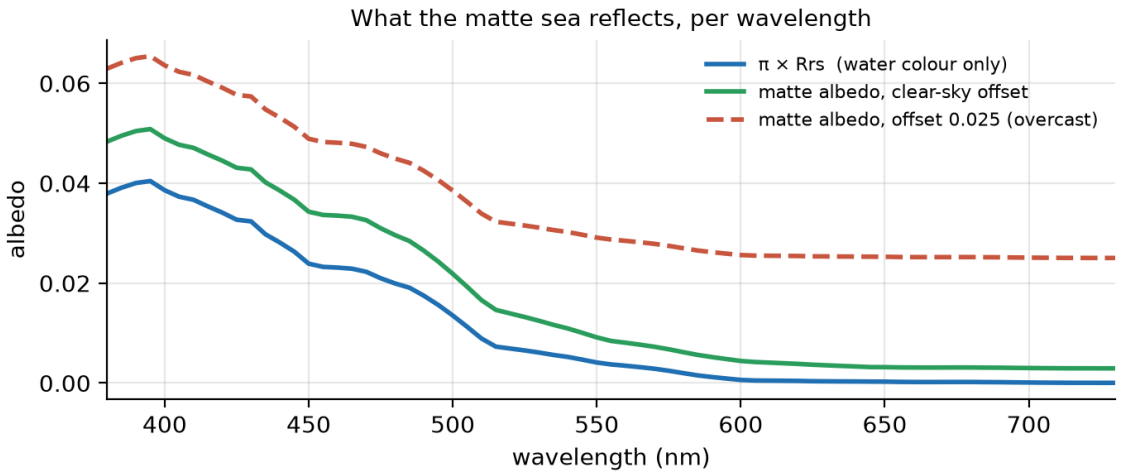


Figure 2. The albedo the matte sea has per wavelength: the measured colour alone (blue), with the clear-sky offsets (green) and with a constant offset of 0.025 (dashed). The blue peak and the dark red are the sea's colour; the offset lifts all of it, relatively most in the red.

3 What it loses

- **The sun glint.** A flat diffuse surface has no mirror. Near the glint the matte sea is far too dark.
- **The sea-surface polarization.** A Lambertian surface depolarizes light, so the polarized signal of the sea surface is lost. I expect this; I did not test it.
- **The wind dependence** and the **angular shape** of the sea's brightness (a real sea is not equally bright in every direction).
- **Whitecaps and foam**, not modelled in either sea model here.

It is meant for the colour of the sea in scenes with clouds. It is not meant for polarization work or for geometries near the glint; use the full models cloud-free for those.

4 How well it works

Cloud-free (the full model can run, so there is a truth). MODIS bands b03 / b04 / b01, offsets fitted at the test-pixel geometry, 3e7 photons:

Geometry (sun zenith, view zenith, azimuth)	Matte / full model
Test pixel: 39°, 13°, off the glint (where the offset was fitted)	0.999 / 1.000 / 1.007
20°, 30°, toward the sun	1.005 / 1.007 / 1.014
50°, 45°, cross-sun	0.979 / 0.975 / 0.975
Test pixel, RGB band set (440 / 550 / 670 nm)	0.986 / 1.003 / 1.012
30°, 30°, **in the sun glint**	0.36 / 0.16 / 0.09

Away from the glint the matte sea stays within about ±2.5% of the full model. In the glint it fails completely, as it must.

Under a plane-parallel cloud (1-2 km, effective radius 10 µm; the full model can still run there). The clear-sky offsets become too small, because the sea now reflects a bright sky:

Cloud optical depth	Matte (clear-sky offset) / full	Offset that would match the full model (b03 / b04 / b01)
0 (clear)	1.002 / 1.004 / 1.009	0.010 / 0.004 / 0.003

Cloud optical depth	Matte (clear-sky offset) / full	Offset that would match the full model (b03 / b04 / b01)
2	0.953 / 0.911 / 0.905	0.021 / 0.024 / 0.022
5	0.962 / 0.955 / 0.941	0.034 / 0.033 / 0.030
20	0.991 / 0.997 / 0.985	not meaningful: the sea is hardly visible

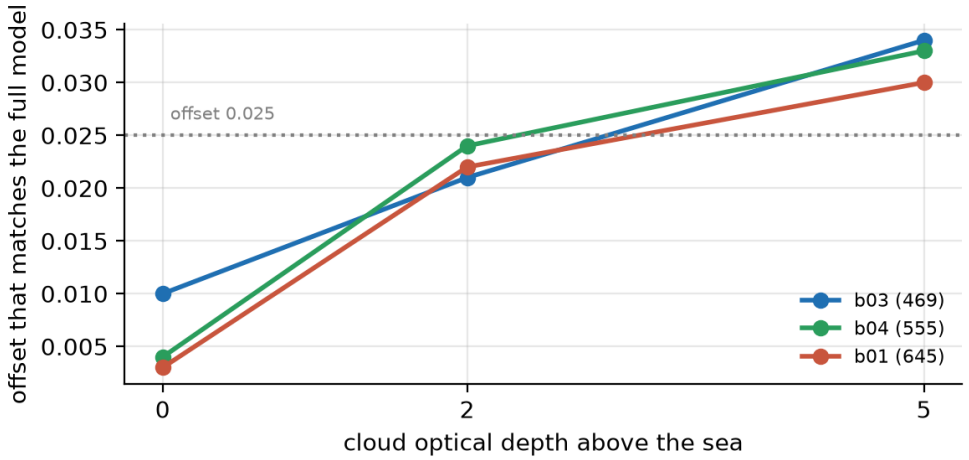


Figure 3. The offset that makes the matte sea equal the full model, against the optical depth of the cloud above. It grows from the clear-sky values to about 0.02-0.035 in every band.

Why one offset cannot fit a cumulus field. It has clear sea between the clouds and sea under or near them. An offset of 0.025 used where the sky is clear would make the sea too bright by about 10% in blue, 33% in green and 66% in red. The default is therefore the clear-sky table, and you may set one number yourself. About 0.02-0.03 suits an overcast sky; choosing it is a judgement the model cannot make for you.

In a 3D cloud scene (bundled cumulus, 16×16 pixels, 5.1e6 photons, MODIS bands) there is no full model to compare with. The table shows what the matte sea adds to a reflecting-only sea. The darkest quarter of the pixels is mostly sea; the mean is mostly cloud. A repeat of the reflecting-only run agrees within 1%.

3D cloud scene, b03 / b04 / b01	Mean over the image	Darkest quarter of the pixels
Sea reflection only	0.160 / 0.126 / 0.118	0.081 / 0.041 / 0.023
Matte sea, clear-sky offsets	0.174 / 0.140 / 0.117	0.098 / 0.045 / 0.025
Matte sea, offset 0.025	0.187 / 0.152 / 0.133	0.106 / 0.056 / 0.037

With the clear-sky offsets the darkest pixels brighten by 21% in blue, 11% in green and 8% in red; with 0.025 by 31%, 38% and 58%. More light, most in the blue, is what the sea's colour should add, but there is no truth here: read it as indicative, not validated. (A first test with 16 times fewer photons was too noisy to read; the same case differed by up to 35% between runs.)

5 How to use it

In a scenario file (or the GUI: Surface = ocean, tick the water body, choose *matte*):

```
"surface": "ocean", "water_body": true, "water_model": "matte",
"sea_spectrum_file": "configs/sea_colour/moby_clear_water_rrs.csv",
"matte_offset": null
```

Setting	Meaning
`water_model`	`"matte"` is the only model that works with a cloud (`chlorophyll` and `measured` are cloud-free only; the configuration check says so).
`sea_spectrum_file`	A two-column CSV (nm, Rrs in 1/sr). Clear water (MOBY) by default; NOMAD files for greener water are in configs/sea_colour/.

Setting	Meaning
`matte_offset`	null: clear-sky table. A number from 0 to 0.1: that offset at every wavelength (GUI: 0 means the table). About 0.02-0.03 for an overcast sky.

Guidance: clear or mostly clear scene, leave the offset empty; heavily overcast scene, try 0.025; mixed cumulus, treat the sea colour as approximate and compare two offsets. Do not use the matte sea near the glint or to study sea-surface polarization.

6 Open points

- Whether the matte sea really loses the sea surface's polarization: expected, untested.
- A remaining gap of about 5% in blue and green between the full model and the MODIS pixel is unexplained and applies to the matte sea too.
- The offset is a single number per image; a per-pixel offset from the local sky brightness would be a possible refinement.
- For the full sea models, the spectra and the units (Rrs against R(0-)), see *SMARTG_sea_model.pdf*.